

# DROS-PGM: A Deterministic Post-Compromise Execution Containment Substrate for Autonomous AI Workloads (v2.0)

Chun-Cheng (Jimmy) Chen  
*Top-Celestial Company Ltd.*  
 Taipei, Taiwan R.O.C.  
 jimmychen@dr-os.io

**Abstract**—In modern autonomous AI agent workloads where agents obtain legitimate credentials and invoke consequential tools, traditional application-layer perimeters suffer from fundamental failure modes. Conventional defenses rely on probabilistic semantic guardrails or coarse-grained operating system sandboxes under the assumption of preventing compromise. However, once internal dialogue or interpreter environments succumb to indirect prompt injection, attackers inevitably inherit valid credentials and execute irreversible physical state changes. This paper presents DROS-PGM, a post-compromise execution containment substrate operating at the binary execution control plane. Architecturally, the C-ABI / FFI boundary manages policy evaluation and principal capability attribution, while OS kernel hooks enforce mandatory authorization checks over explicitly instrumented operation classes  $X_{\text{covered}}$ . PGM decouples application principal identity from binary execution authority, maintaining the formal containment invariant via sub-microsecond ( $P50 = 353$  ns) lock-free evaluation and atomic RCU state-pointer swaps. To rigorously evaluate boundary robustness without self-witness circularity, we introduce the PGM-VEP Five-Tier Progressive Falsification Methodology (V1–V5), encompassing attack-equivalent baselines, adaptive white-box state search, negative control meta-verification (5/5 injected flaw detection and 100/100 mutant kill score), a four-stage decoupled ground-truth oracle pipeline ( $O_I \rightarrow O_A \rightarrow O_E \rightarrow O_P$ ), and cross-environment replication across Linux x86\_64, ARM64, and Windows. Across 118,355 total executions (68,355 adversarial + 50,000 benign, BFDR = 0/50,000), zero unauthorized executions or state drifts were observed within explicitly instrumented boundaries. We report these guarantees as empirical invariants over the evaluated state space and establish an open counterexample registry to support ongoing adversarial falsification.

**Index Terms**—Execution Containment Substrate, Post-Compromise Security, Kernel Enforcement Boundary, Dynamic Capability Revocation, Progressive Adversarial Falsification, Decoupled Oracles.

## I. PROBLEM DEFINITION: COMPROMISE DOES NOT ENTAIL AUTHORITY

Autonomous AI Agent architectures routinely operate with elevated privileges. In these environments, security architectures encounter a fundamental Semantic-Kernel Paradox:

- 1) **Semantic Application Middleware (High Semantics, Zero Determinism):** Prompt firewalls and JSON

schema validators operate purely at the natural language plane, lacking physical execution containment.

- 2) **Operating System Sandboxes (High Determinism, Zero Semantics):** Traditional kernel mechanisms enforce binary constraints but lack application semantic context, failing to distinguish between legitimate and compromised agents within the same process.

We formalize the core research thesis: **Compromise of an application principal does not entail inheritance of unrestricted execution authority** ( $C_A \not\Rightarrow C_E$ ).

## II. DUAL BOUNDARIES & ARCHITECTURAL PLACEMENT

The system establishes a strict separation of concerns via a Dual-Boundary Model:

- **Boundary 1 (C-ABI / FFI Attribution Boundary):** Manages policy evaluation and principal capability attribution, evaluating in 353 ns ( $P50$ ) via  $O(1)$  SIMD bitmasks.
- **Boundary 2 (OS Kernel Mandatory Enforcement Boundary):** Intercepts operations over the explicitly instrumented operation classes  $X_{\text{covered}}$ .

$$X_{\text{covered}} = X_{\text{fs}} \cup X_{\text{proc}} \cup X_{\text{net}} \cup X_{\text{ipc}} \quad (1)$$

*Non-Claim Clause:* PGM does not claim syscall-complete mediation over uninstrumented OS kernel paths.

## III. FORMAL INVARIANTS & SECURITY CLAIM MATRIX

### A. Formal Invariants

- 1) **Non-Inheritance of Authority Axiom:**  $C_A \not\Rightarrow C_E$
- 2) **Covered Containment Invariant:**

$$\forall x \in X_{\text{covered}}, \quad \text{Auth}_E(x) = \text{DENY} \implies \text{Exec}_{\text{unauth}}(x) = 0 \wedge \Delta S_{\text{Sobs}} = 0 \quad (2)$$

- 3) **Ghost Syscall Measurement Oracle:**

$$G = \#\{x \in X_{\text{covered}} \mid \text{Auth}_E(x) = \text{DENY} \wedge \text{SyscallObserved}(x) = 1\} \quad (3)$$

- 4) **Linearization Semantics of Revocation:** Dynamic revocation takes effect deterministically for all subsequent calls at  $T_{\text{swap}} = 420$  ns.

Protected under U.S. Provisional Patent Application No. 64/111,973 (Patent Pending). Permanent Academic Citation: Zenodo Record DOI: 10.5281/zenodo.21903687. Open-Source Testbed: <https://github.com/Top-Celestial-Company-Ltd/DROS-VEP-lite>.

TABLE I  
DROS-PGM SECURITY CLAIM BOUNDARY MATRIX

| Claim Category     | Statement & Bounded Scope                                                 | Epistemic Status        |
|--------------------|---------------------------------------------------------------------------|-------------------------|
| <b>Claimed</b>     | In $X_{\text{covered}}$ , DENY operations produce zero unauthorized execs | Architectural Invariant |
| <b>Claimed</b>     | Revocation takes effect immediately at $T_{\text{swap}}$                  | Linearization Semantics |
| <b>Observed</b>    | Observed ghost syscall count $G = 0$                                      | Empirical Fact          |
| <b>Observed</b>    | Observed state drift $\Delta S = 0$ over $\mathcal{S}_{\text{obs}}$       | Multi-Oracle Fact       |
| <b>Not Claimed</b> | Global syscall-complete mediation                                         | Explicit Non-Claim      |
| <b>Not Claimed</b> | Defense against kernel rootkits / MMU bypass                              | Host TCB Assumption     |
| <b>Not Claimed</b> | Upstream semantic bitmask correctness                                     | Upstream Duty           |

#### IV. THREAT MODEL: POST-COMPROMISE ASSUMPTION ( $C_A = 1$ )

- **In-Scope Attacker:** Full application compromise ( $C_A = 1$ ), in-memory credential access, multi-agent confused deputy evasion.
- **TCB Assumptions:** PGM binary gate, CPU Ring 0/MMU, OS LSM drivers, immutable audit logs.

#### V. PGM-VEP VALIDATION METHODOLOGY (V1–V5)

Enforcing the No-Self-Witness Principle ( $O_I \neq O_A \neq O_E \neq O_P$ ), we structure evaluation across five tiers:

TABLE II  
PGM-VEP METHODOLOGY TIERS TO ASSET MAPPING

| Methodology Tier      | Open-Source Executable Harness | Evidence Asset Path     |
|-----------------------|--------------------------------|-------------------------|
| <b>V1 (Baseline)</b>  | run_benchmark.py               | CYBERMES_REPORT.md      |
| <b>V2 (Search)</b>    | stress_test_accelerator.py     | BOSSy24H_Report.md      |
| <b>V3 (Meta-Test)</b> | conformance_test.py            | conformance_report.json |
| <b>V4 (Oracle)</b>    | replay.py                      | COMPARATIVE_REPORT.md   |
| <b>V5 (Cross-Env)</b> | docker-compose.yml             | benchmark_summary.json  |

#### VI. EXPERIMENTAL RESULTS & METRIC ANALYSIS

##### A. Evaluation Accounting Matrix

##### B. Meta-Verification (V3)

The harness killed 100/100 mutants (Mutation Score = 1.0) and caught 5/5 sabotaged binaries, confirming discriminator sensitivity.

##### C. Performance & Selected SPEC CPU2017 Breakdown

Geometric mean overhead on Intel Xeon Gold 6330 was **+1.20% ( $\pm 0.12\%$ )**. Across a 72-hour soak test, PGM sustained an aggregate throughput of **952,991 QPS across 16 concurrent worker threads**.

#### VII. RELATED WORK & TAXONOMY

*Note: To the best of our survey across publicly documented literature, we did not identify an existing system reporting all five validation dimensions under a unified open package.*

TABLE III  
EVALUATION ACCOUNTING MATRIX

| Tier                 | Configuration & Architecture        | Total Execs    | CE       | Status          |
|----------------------|-------------------------------------|----------------|----------|-----------------|
| <b>V1</b>            | B0/B1 Baseline Equivalence          | 300            | 0        | <b>PASS</b>     |
| <b>V2</b>            | White-box search (21.4M Tokens)     | 60,000         | 0        | <b>PASS</b>     |
| <b>V3</b>            | 5 sabotaged binaries + 100 mutants  | 105            | 0        | <b>PASS</b>     |
| <b>V4</b>            | 4-stage fact chain + 9 vectors      | 450            | 0        | <b>PASS</b>     |
| <b>V5</b>            | 3 OS environment profiles           | 7,500          | 0        | <b>PASS</b>     |
| <b>Adv. Subtotal</b> | <b>V1–V5 Cumulative Adversarial</b> | <b>68,355</b>  | <b>0</b> | <b>PASS</b>     |
| <b>Benign</b>        | Selected SPEC CPU2017 + Tools       | 50,000         | 0        | <b>BFDR=0</b>   |
| <b>Total</b>         | <b>Full-Spectrum Comprehensive</b>  | <b>118,355</b> | <b>0</b> | <b>ALL PASS</b> |

TABLE IV  
PGM IN-BAND LATENCY DISTRIBUTION ( $10^7$  SAMPLES, UNIT: NS)

| Metric                                         | Min | P50        | P99 | Max | 95% CI     |
|------------------------------------------------|-----|------------|-----|-----|------------|
| <b>C-ABI Policy Lookup</b>                     | 312 | <b>353</b> | 412 | 620 | [351, 355] |
| <b>RCU Swap (<math>T_{\text{swap}}</math>)</b> | 390 | <b>420</b> | 438 | 580 | [416, 424] |
| <b>Fail-Closed Denial</b>                      | 410 | <b>465</b> | 488 | 640 | [462, 468] |

#### VIII. CONCLUSION & DECLARATIONS

This paper presented DROS-PGM, establishing a deterministic binary execution containment boundary evaluated across 18,355 executions with zero counterexamples.

#### ACKNOWLEDGMENT & AI COLLABORATION DISCLOSURE

In accordance with IEEE / ACM 2023+ guidelines:

- 1) **Research Originality & IP:** The system architecture, dual boundaries, formal invariants, V1–V5 methodology, and patent claims (U.S. PPA No. 64/111,973) were independently conceived and evaluated by Chun-Cheng (Jimmy) Chen.
- 2) **Role of AI Tools:** LLMs were utilized strictly as assistive productivity tools for proofreading, LaTeX typeset-

TABLE V  
METHODOLOGICAL RIGOR MATRIX ACROSS DEFENSE SYSTEMS

| Defense Category            | Attack-Equiv | Meta-Verify          | Decoupled Oracle |
|-----------------------------|--------------|----------------------|------------------|
| Prompt Guardrails           | Weak         | None                 | None             |
| MCP PEP Gateways            | Partial      | None                 | Weak             |
| Traditional OS Sandbox      | Feasible     | None                 | Weak             |
| Commercial Endpoint         | Internal     | Undisclosed          | Undisclosed      |
| Capability Kernel           | Yes          | Limited              | Partial          |
| <b>DROS-PGM (This Work)</b> | <b>Yes</b>   | <b>Yes (100/100)</b> | <b>Yes</b>       |

ting, and formatting. The author retains full intellectual and legal responsibility.

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